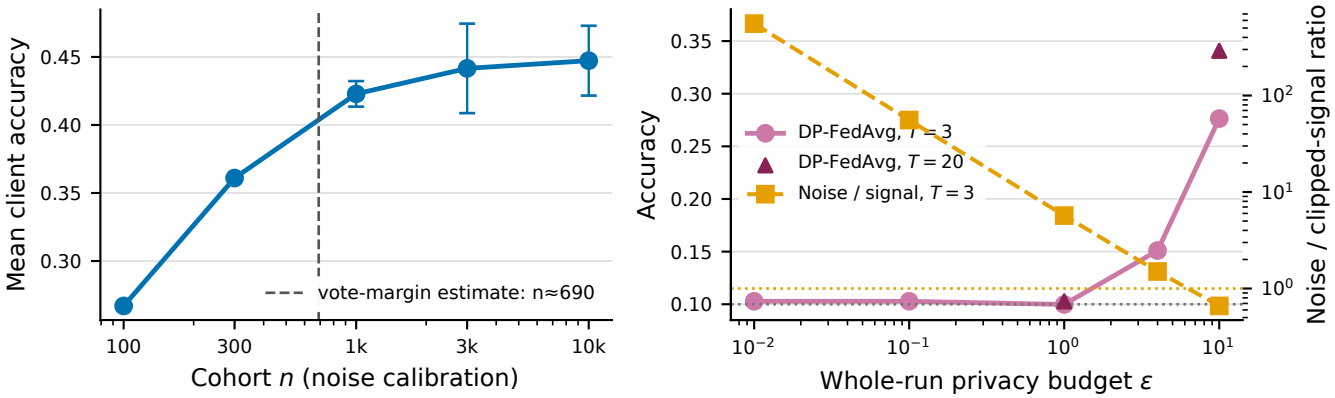
(a) Consensus operating scale
Mean client accuracy
0.45
0.40
0.35
0.30
vote-margin estimate: n≈690
100
300
1k
3k
10k
Cohort n (noise calibration)
(b) Model-update noise scale
Accuracy
0.35
0.30
0.25
0.20
0.15
0.10
DP-FedAvg, $T = 3$
DP-FedAvg, $T = 20$
Noise / signal, $T = 3$
10^{-2}
10^{-1}
10^{0}
10^{1}
Whole-run privacy budget ε
Noise / clipped-signal ratio
10^{2}
10^{1}
10^{0}